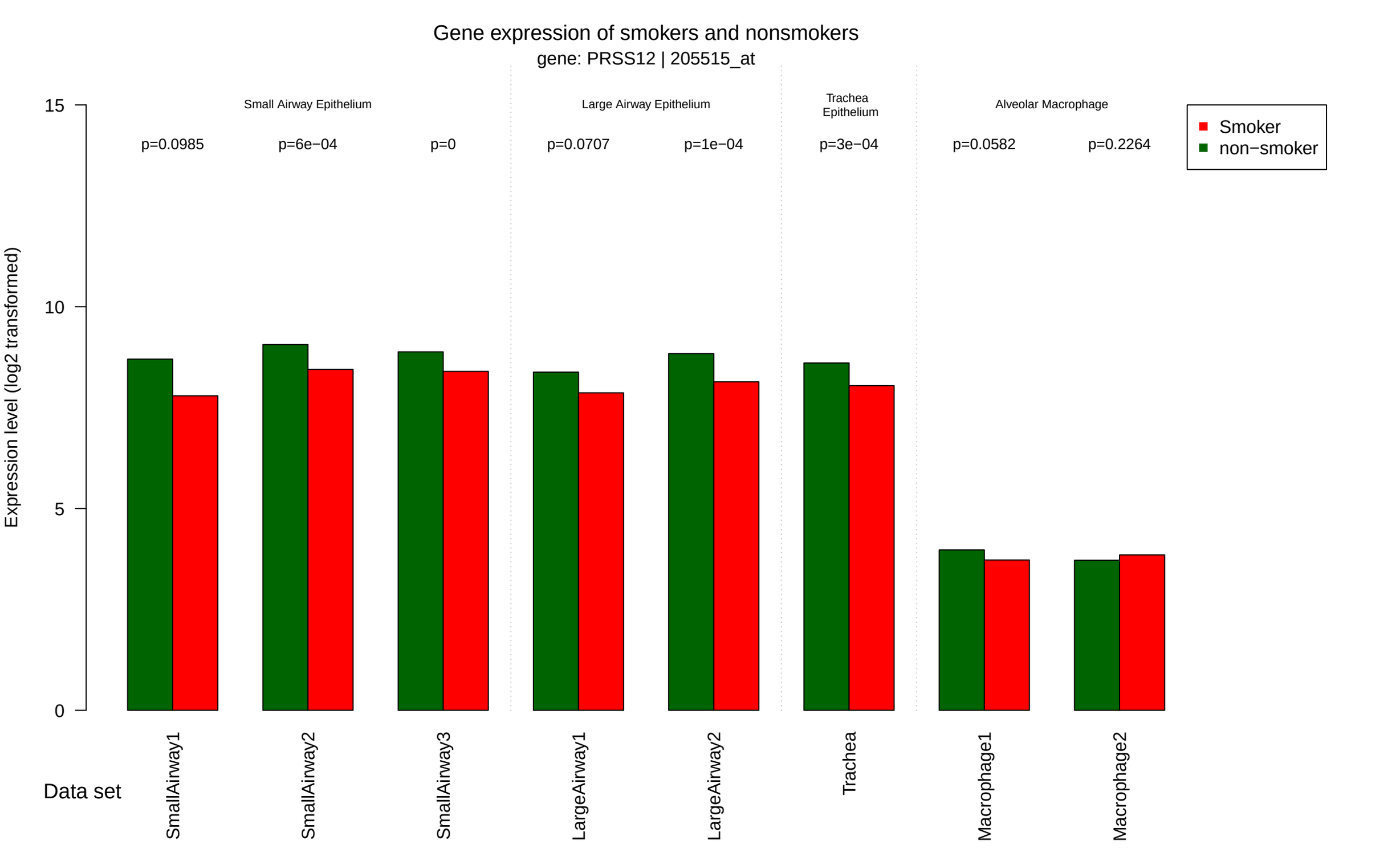
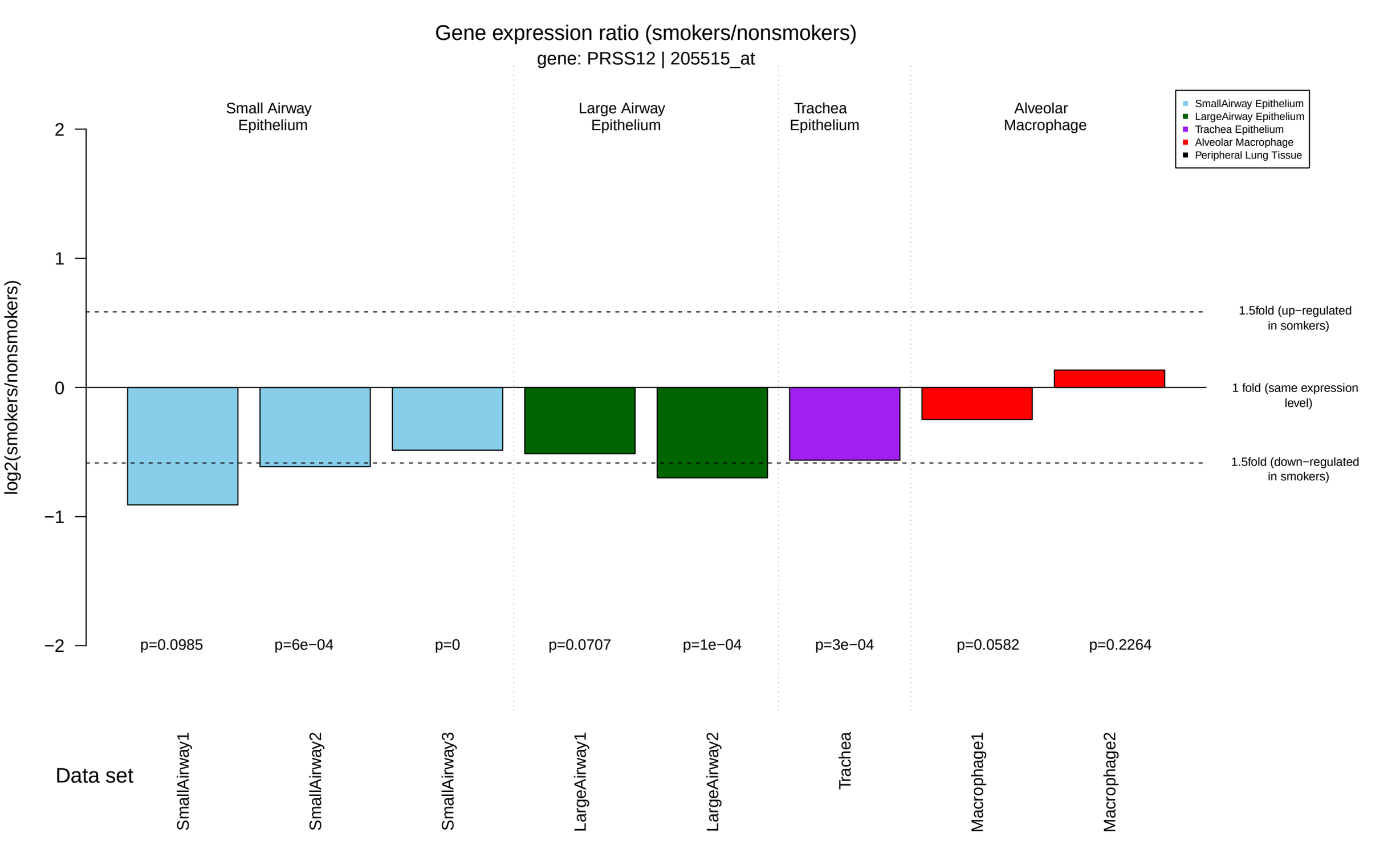
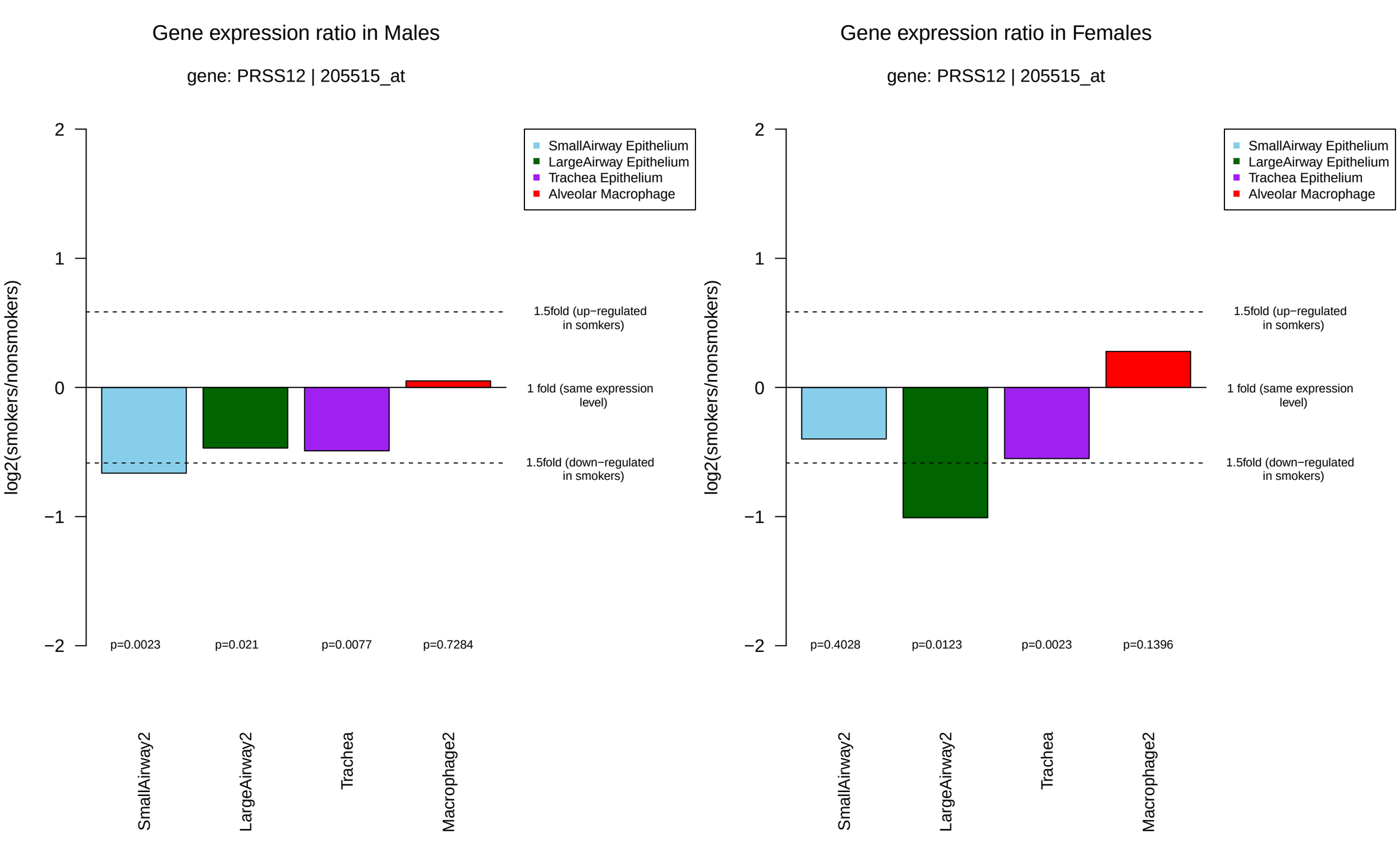
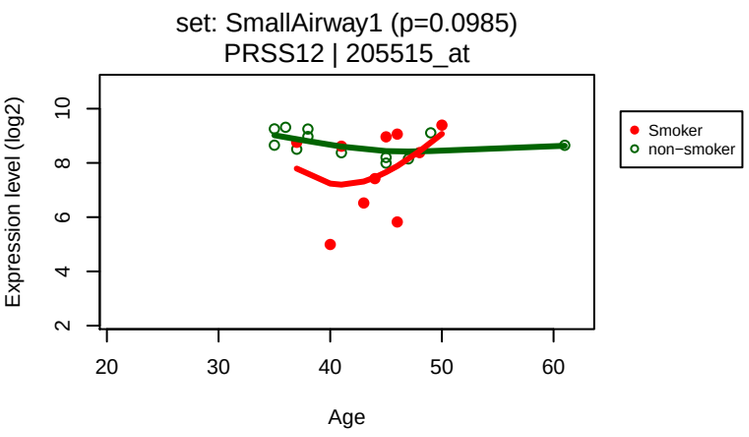
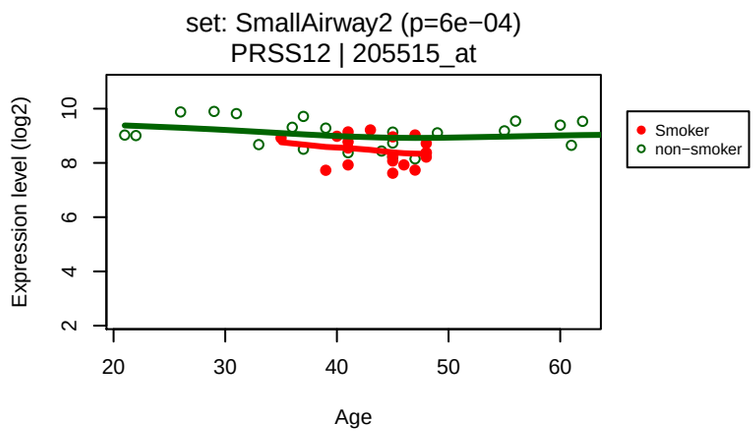
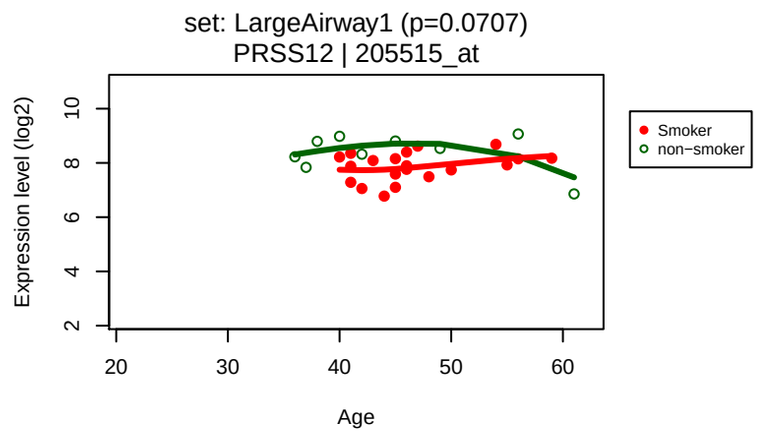

Gene expression of smokers and nonsmokers
gene: PRSS12 | 205515_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0985
p=6e−04
p=0
p=0.0707
p=1e−04
p=3e−04
p=0.0582
p=0.2264
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

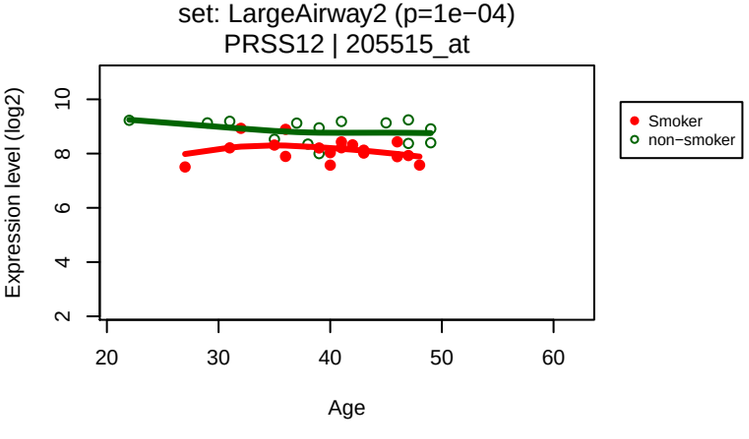

Gene expression ratio (smokers/nonsmokers)
gene: PRSS12 | 205515_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0985
p=6e−04
p=0
p=0.0707
p=1e−04
p=3e−04
p=0.0582
p=0.2264
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

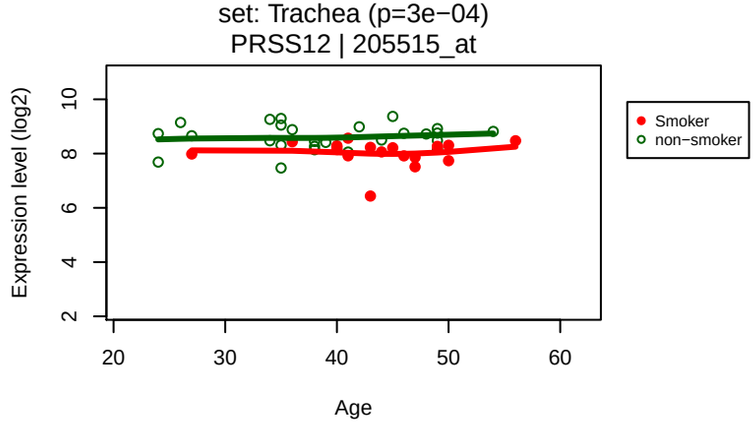

Gene expression ratio in Males
gene: PRSS12 | 205515_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0023
p=0.021
p=0.0077
p=0.7284
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS12 | 205515_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4028
p=0.0123
p=0.0023
p=0.1396
SmallAirway2
LargeAirway2
Trachea
Macrophage2

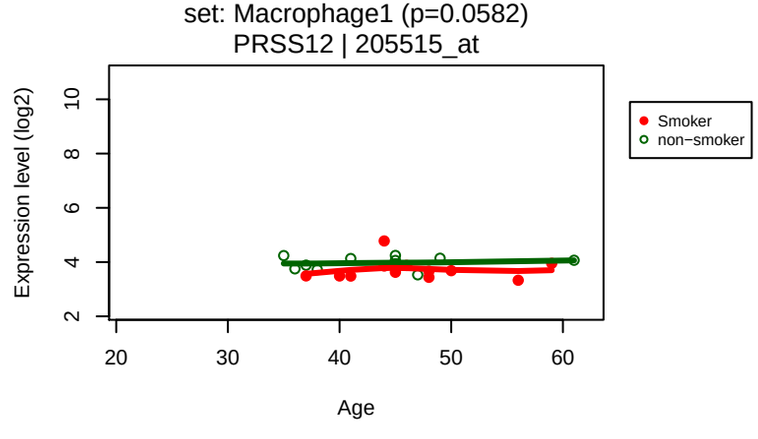

set: SmallAirway1 (p=0.0985)
PRSS12 | 205515_at
Expression level (log2)
Age
Smoker
non−smoker

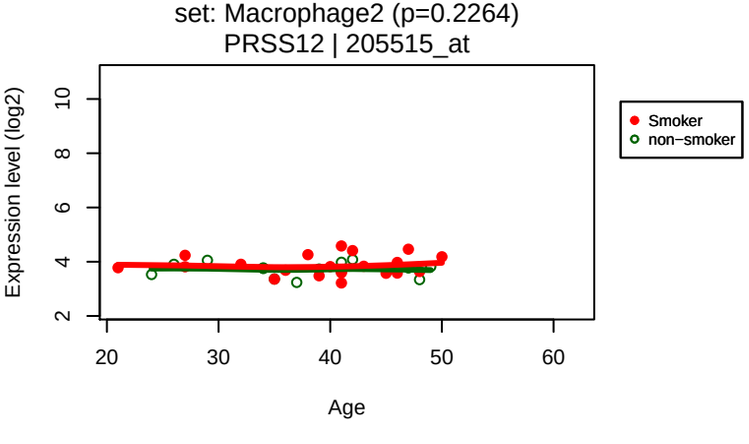

set: SmallAirway2 (p=6e−04)
PRSS12 | 205515_at
Expression level (log2)
Age
Smoker
non−smoker

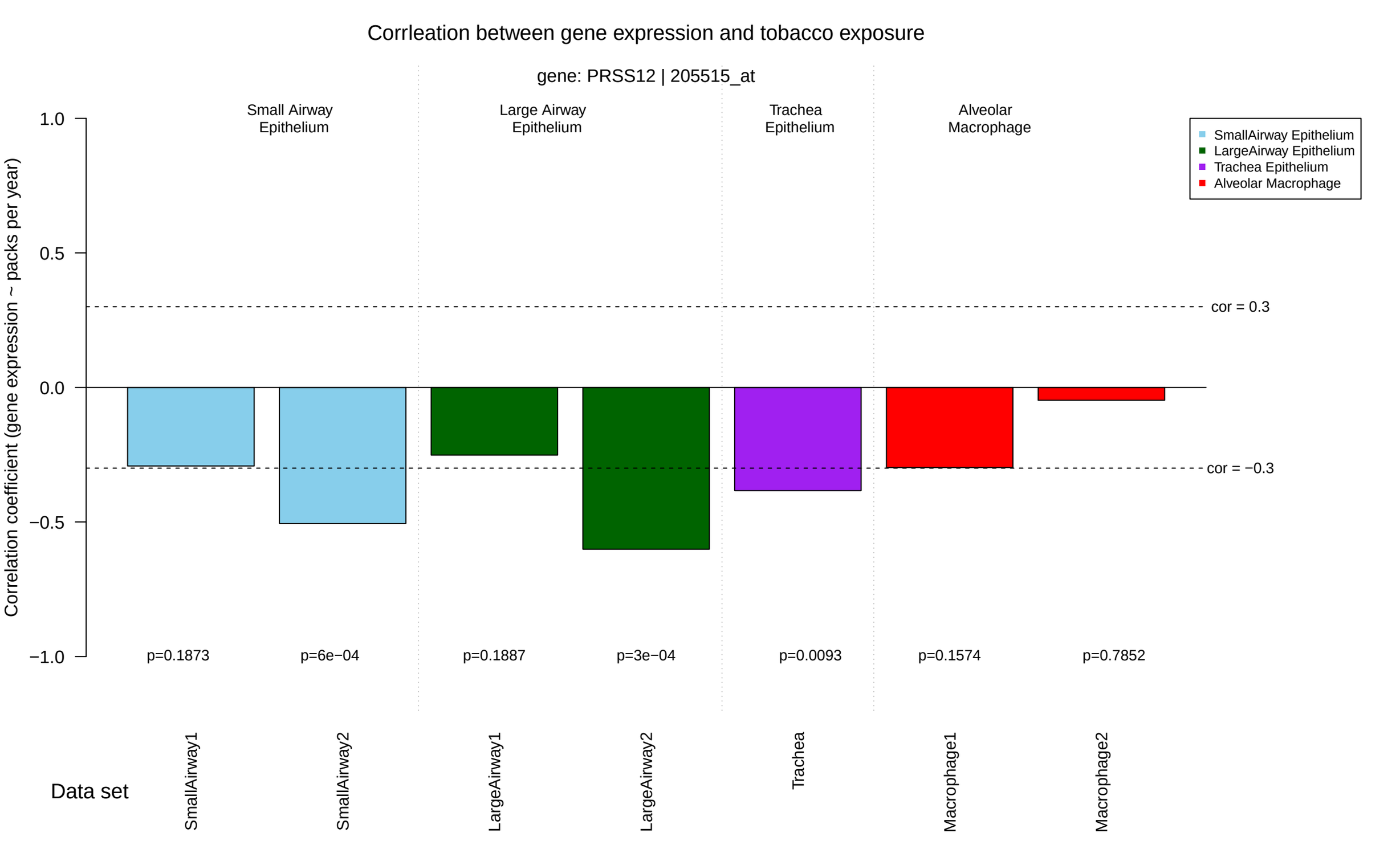

set: LargeAirway1 (p=0.0707)
PRSS12 | 205515_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=1e−04)
PRSS12 | 205515_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
PRSS12 | 205515_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0582)
PRSS12 | 205515_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2264)
PRSS12 | 205515_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS12 | 205515_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1873
p=6e−04
p=0.1887
p=3e−04
p=0.0093
p=0.1574
p=0.7852
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2